

CSE 370– Database Systems
Assignment 2
Spring 2025

Submission

Instructions:

1. Write your name, id, section on top of the first page
2. Your answer should be handwritten, take pictures and create a single pdf.
3. **Submit the pdf in the following form:** <https://forms.gle/hmKaSc46RMXoGHpv5>
4. **Submission Deadline:** 15th May 2025 (Thursday), 11:59 pm (midnight)

5. NO LATE SUBMISSION WILL BE ACCEPTED

Question 1

[4 X 2 =8 Marks]

X	Y	Z	A
abcefg	1	q	10
xyz	2	p	11
feg	3	q	12
xyz	2	p	13
abcdefg	3	q	10

State which of the following dependencies are valid and which are not. For each dependency, briefly write the reasons. The first one is explained and solved for you:

A. $XY \rightarrow A$

Functional dependency is valid if for the same value of XY, the corresponding value of A is also the same. Here (“xyz”, 2) is the repeated value for X and Y in row 2 and 4. The corresponding value in A is 11 and 13 respectively, therefore this functional dependency is not valid according to the definition of FD.

[Note: if all values of XY were unique, i.e. no repetition, then the FD will be valid as it does not contradict the definition of FD]

B. $X \rightarrow YZ$

C. $A \rightarrow XYZ$

D. $YZ \rightarrow X$

E. $Y \rightarrow Z$

Question 2

[2+3+3 = 8 Marks]

Consider the following relation:

Pokemon_Battles (Battle_id, Trainer_id, Battle_name, Trainer_Name, battle_date, {rewards}, Pokemon_id, pokemon_name, type, weakness, score)

Note: Multivalued is shown in curly braces {}. The primary key of the relation is underlined.

The relation has the following additional functional dependencies:

FD1: $\text{Battle_id} \rightarrow \text{Battle_name}, \text{battle_date}$

FD2: $\text{Trainer_id} \rightarrow \text{Trainer_Name}$

FD3: $\text{Pokemon_id} \rightarrow \text{pokemon_Name}, \text{type}, \text{weakness}$

FD4: $\text{type} \rightarrow \text{weakness}$

a. Explain if the above relation is in first normal form (1NF) or not? If not, apply 1NF normalization.

b. Explain if the schema in (a) is in the second normal form (2NF) or not? If not, apply 2NF normalization.

c. Explain if the schema in (b) is in the third normal form (3NF) or not? If not, apply 3NF normalization.

Show full schema after each normalization step.